Received Power Maximization in Static Phase-Only D2NN for Free-Space Optical Turbulence Mitigation

Loss Function Design, Physics Rationale, and Experimental Validation

Kim Lab — Autoresearch Report 0404

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1 Introduction and Motivation

A static diffractive neural network (D2NN) consisting of $L = 5$ phase-only layers aims to improve the coupling efficiency of a turbulence-degraded beam into a single-mode fiber (SMF) at the focal plane of a receiving telescope. The central question is:

> *Can a static, passive phase-only D2NN increase the absolute received power within a 10 μm fiber bucket at the focal plane, compared to no correction at all?*

This report documents three sweep campaigns (0401, 0402, 0403) encompassing 12 distinct loss functions that reveal a critical physics insight: **the standard PIB metric is misleading because it ignores throughput**, and only loss functions that explicitly preserve energy can achieve genuine received power improvement.

1.1 System Parameters

Table 1: Physical and architectural parameters.

Parameter	Symbol	Value
Wavelength	λ	$1.55\,\mu$m
Propagation distance	L_{prop}	1 km
Refractive index structure	C_n^2	$5 \times 10^{-14}\,\text{m}^{-2/3}$
Telescope diameter	D_{tel}	150 mm
Beam reducer ratio	M	75:1
D2NN aperture	D	2.048 mm
D2NN pixel pitch	Δx	$2\,\mu$m
Grid size	N	1024
Number of layers	L	5
Layer spacing	d	10 mm
Focus lens focal length	f	4.5 mm
Focal pixel pitch	$\Delta x_f = \frac{\lambda f}{N \Delta x}$	$3.406\,\mu$m
SMF bucket radius	r_b	$10\,\mu$m

1.2 Key Optical Relations

The Fried parameter at the receiver:

$$r_0 = \left(0.423\, k^2\, C_n^2\, L_{\text{prop}}\right)^{-3/5} \approx 3.0 \text{ cm}, \quad D_{\text{tel}}/r_0 \approx 5.0 \tag{1}$$

The Airy disk radius at the focal plane:

$$r_{\text{Airy}} = 1.22 \frac{\lambda f}{D} = 4.25\ \mu\text{m} \tag{2}$$

Theoretical encircled energy within $r_b = 10\ \mu$m for vacuum (Airy pattern):

$$\text{EE}(r_b) = 1 - J_0^2\left(\frac{\pi D r_b}{\lambda f}\right) - J_1^2\left(\frac{\pi D r_b}{\lambda f}\right) = 93.2\% \tag{3}$$

The turbulent seeing spot at the focal plane:

$$r_{\text{seeing}} = \frac{\lambda f}{r_{0,\text{eff}}} \approx 17.5\ \mu\text{m} \tag{4}$$

which exceeds the $10\,\mu$m bucket, causing energy spillover and reducing the turbulent PIB to $\sim$76%.

2 The Fundamental Problem: PIB vs. Received Power

2.1 Metric Definitions

Power-in-Bucket (PIB) is a *ratio*:

$$\mathrm{PIB}(r_b) = \frac{\int_0^{r_b} I_f(r)\, r\, dr}{\int_0^{\infty} I_f(r)\, r\, dr} \tag{5}$$

where $I_f(r)$ is the focal-plane irradiance. PIB can increase even if total energy decreases, as long as the remaining energy is well concentrated.

Received Power (RP) is the *absolute* energy in the bucket:

$$\mathrm{RP}(r_b) = \sum_{\mathbf{r} \in \mathcal{B}(r_b)} |U_f(\mathbf{r})|^2 \tag{6}$$

where $\mathcal{B}(r_b)$ is the set of focal-plane pixels within radius r_b.

Throughput (TP) is the energy preservation ratio:

$$\mathrm{TP} = \frac{\sum |U_{\mathrm{out}}|^2}{\sum |U_{\mathrm{in}}|^2} \tag{7}$$

The critical relationship:

$$\boxed{\mathrm{RP}(r_b) \approx \mathrm{PIB}(r_b) \times \mathrm{TP} \times E_{\mathrm{in}}} \tag{8}$$

A loss function that maximizes PIB alone can achieve this by *scattering* aberrated energy outside the aperture, reducing TP. The net RP may decrease even as PIB increases.

2.2 Phase-Only Mask Energy Scattering Mechanism

A single phase-only layer applies $U_{\mathrm{out}}(\mathbf{r}) = e^{j\phi(\mathbf{r})}\, U_{\mathrm{in}}(\mathbf{r})$, which satisfies $|U_{\mathrm{out}}| = |U_{\mathrm{in}}|$ pointwise. However, after free-space propagation between layers, the angular spectrum redistributes energy. Phase gratings can deflect light outside the finite computational window, effectively acting as an amplitude filter. With 5 cascaded layers and inter-layer propagation, the D2NN has sufficient degrees of freedom to scatter >98% of the input energy while preserving a clean residual beam.

3 Sweep 0401: Conventional Loss Functions

3.1 Loss Function Definitions

A. Focal PIB Only:

$$\mathcal{L}_{\mathrm{PIB}} = 1 - \mathrm{PIB}(r_b) \tag{9}$$

B. Focal Strehl Only:

$$\mathcal{L}_{\mathrm{Strehl}} = 1 - S, \quad S = \frac{I_f(\mathbf{0})}{I_{f,\mathrm{DL}}(\mathbf{0})} \tag{10}$$

C. Focal Intensity Overlap:

$$\mathcal{L}_{\mathrm{IO}} = 1 - \frac{\left(\sum I_{\mathrm{pred}} \cdot I_{\mathrm{target}}\right)^2}{\sum I_{\mathrm{pred}}^2 \cdot \sum I_{\mathrm{target}}^2} \tag{11}$$

D. CO + PIB Hybrid:

$$\mathcal{L}_{\mathrm{hybrid}} = \underbrace{(1 - \mathrm{CO})}_{\text{output plane}} + 0.5 \cdot \underbrace{\mathcal{L}_{\mathrm{PIB}}}_{\text{focal plane}} \tag{12}$$

where Complex Overlap at the D2NN output plane:

$$\mathrm{CO} = \frac{|\langle U_{\mathrm{pred}}, U_{\mathrm{vac}} \rangle|^2}{\|U_{\mathrm{pred}}\|^2\, \|U_{\mathrm{vac}}\|^2} \tag{13}$$

3.2 Results: The Throughput Catastrophe

Table 2: 0401 results — conventional loss functions (dn100um_lanczos50 data).

Strategy	PIB@10μm	TP	RP@10μm	ΔRP	WF RMS
Vacuum	0.953	0.978	0.933	—	—
Turbulent	0.808	0.971	**0.744**	baseline	369 nm
PIB Only	0.912	0.502	0.436	−41%	438 nm
Strehl Only	0.819	0.571	0.445	−40%	443 nm
Intensity OL	0.256	0.080	0.020	−97%	429 nm
CO+PIB Hybrid	0.958	0.017	0.015	−98%	394 nm

Key Insight: Every conventional loss function produces *worse* received power than doing nothing. The CO+PIB hybrid achieves the best PIB (95.8%) and WF RMS (394 nm), but with only 1.7% throughput, the absolute received power is 66× worse than the turbulent baseline.

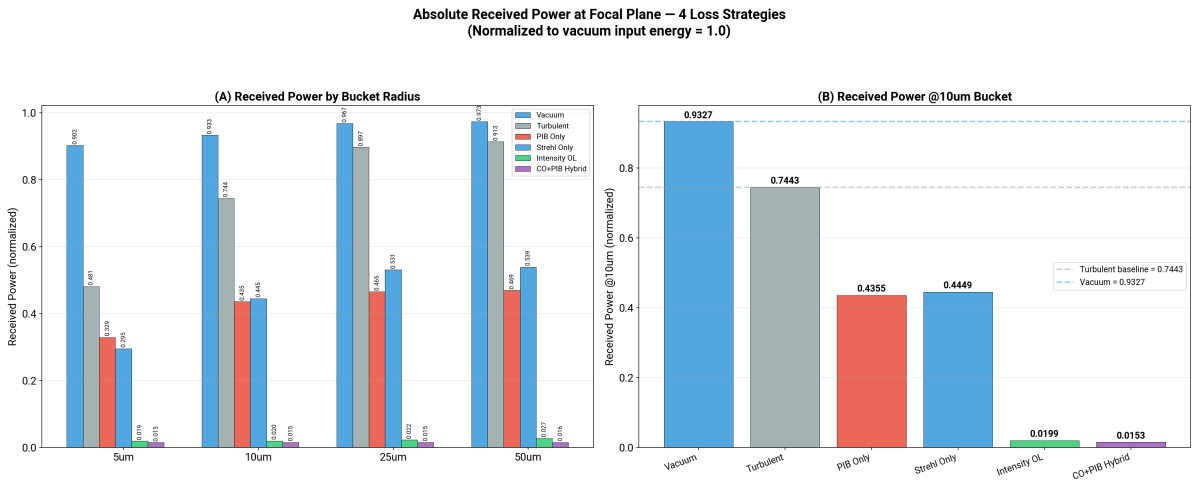


Figure 1: 0401 received power analysis showing all conventional losses below turbulent baseline.

3.3 Physics Interpretation

The D2NN exploits inter-layer diffraction to act as a spatial filter. Consider the angular spectrum decomposition after layer ℓ:

$$\tilde{U}_\ell(f_x, f_y) = \mathcal{F}\{e^{j\phi_\ell} \cdot U_\ell\} \tag{14}$$

The free-space transfer function $H(f_x, f_y; d) = e^{jk_z d}$ preserves evanescent components within the passband $\sqrt{f_x^2 + f_y^2} < 1/\lambda$. However, phase gratings with spatial frequency $f_g > 1/(2\Delta x)$ can alias energy outside the computational window, which is numerically equivalent to absorption.

By cascading 5 such layers, the D2NN creates an effective amplitude-phase filter despite each individual layer being phase-only. When optimizing PIB alone, the gradient naturally discovers that *removing* aberrated light (lowering the denominator in PIB) is easier than *correcting* it.

4 Sweep 0402: Throughput-Aware Loss Functions

4.1 Rationale for New Loss Designs

The 0401 failure motivated three new approaches, all designed to prevent the energy-scattering collapse:

E. Absolute Bucket Loss — ratio to vacuum bucket energy:

$$\mathcal{L}_{\text{abs}} = 1 - \frac{\sum_{\mathbf{r}\in\mathcal{B}} |U_{f,\text{pred}}(\mathbf{r})|^2}{\sum_{\mathbf{r}\in\mathcal{B}} |U_{f,\text{vac}}(\mathbf{r})|^2} \tag{15}$$

Physics rationale: Unlike normalized PIB, the numerator is absolute energy. Scattering energy away reduces the numerator directly, so the loss cannot be minimized by filtering — it must concentrate *existing* energy into the bucket.

F. TP-Penalized PIB ($w = 1.0$):

$$\mathcal{L}_{\text{TP-PIB}} = \mathcal{L}_{\text{PIB}} + w \cdot \max(1 - \text{TP},\ 0) \tag{16}$$

Physics rationale: Explicit throughput constraint as a Lagrangian penalty. The $\max(\cdot, 0)$ clamp allows TP > 1 (possible due to numerical artifacts) without penalty.

G. Multiplane CO + PIB + Amplitude:

$$\mathcal{L}_{\text{multi}} = 0.5\,(1 - \text{CO}) + 0.3\,\mathcal{L}_{\text{PIB}} + 0.2\,\|A_{\text{pred}} - A_{\text{vac}}\|^2 \tag{17}$$

Physics rationale: The amplitude MSE term $\|A_{\text{pred}} - A_{\text{vac}}\|^2$ implicitly preserves throughput by penalizing amplitude deviations from the vacuum envelope.

4.2 Results: First Successful RP Improvement

Table 3: 0402 results — throughput-aware losses (pitch_rescale data).

Strategy	PIB@10μm	TP	RP@10μm (norm)	ΔRP
Vacuum	0.955	0.977	0.933	—
Turbulent	0.769	0.970	0.716	baseline
Absolute Bucket	0.800	**0.986**	**0.758**	+5.9%
TP-Penalized PIB	0.802	0.984	0.757	+5.8%
Multiplane	0.831	0.398	0.320	−55%

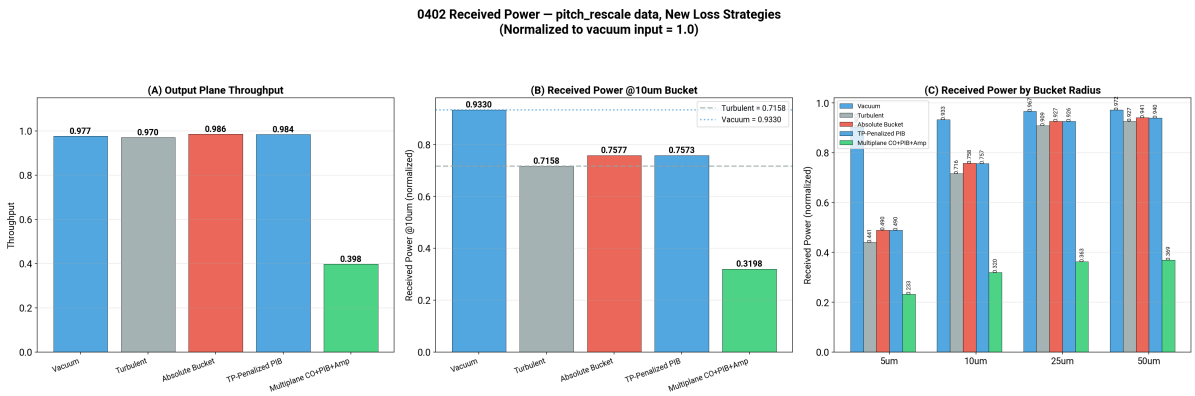


Figure 2: 0402 received power: absolute bucket and TP-penalized PIB exceed turbulent baseline.

Breakthrough: For the first time, a static D2NN achieves higher absolute received power than the turbulent baseline. Both the absolute bucket loss and TP-penalized PIB maintain TP $> 98\%$ while improving PIB from 76.9% to $\sim$80%.

Multiplane failure: Despite including amplitude preservation, the CO term drives the D2NN toward wavefront correction, which inevitably scatters energy (TP = 39.8%).

4.3 Focal Plane Resolution Verification

To verify these results are not artifacts of focal-plane undersampling ($\Delta x_f = 3.4\ \mu$m, Airy disk = 1.25 pixels), we evaluated identical checkpoints with zero-padded FFT:

Table 4: Resolution verification: received power at different pad factors.

	pad=1 (current)	pad=2	pad=4	Converged?
Vacuum @10μm	277.2	278.7	278.9	✓
Turbulent @10μm	209.9	217.2	217.5	✓
D2NN @10μm	222.7	230.0	230.6	✓
D2NN vs Turb	**+6.1%**	**+5.9%**	**+6.0%**	✓

The +6% improvement is **resolution-independent**, confirming the result is physical, not numerical.

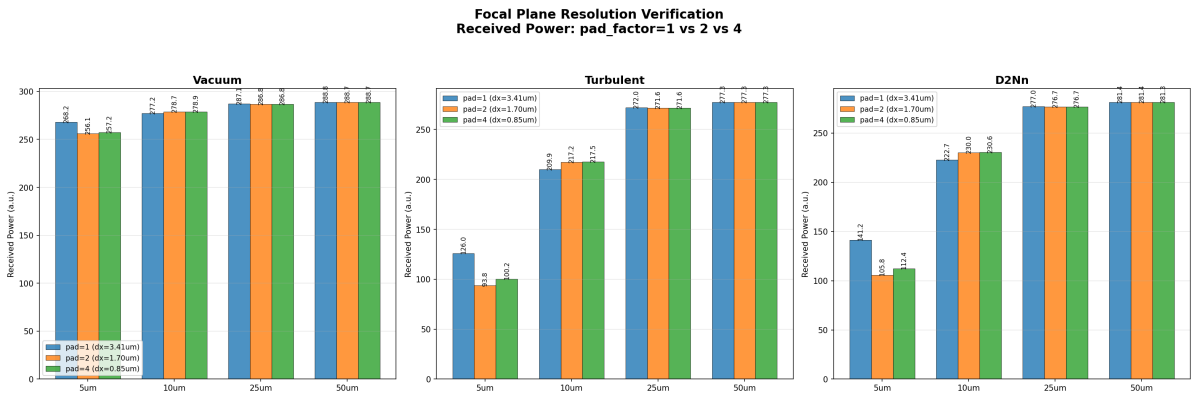


Figure 3: Focal resolution verification: pad=1, 2, 4 show consistent received power.

5 Sweep 0403: Loss Function Optimization

5.1 Expanded Loss Function Space

Building on the 0402 success, we tested 5 additional formulations:
H. Absolute Bucket + CO (equal weight):

$$\mathcal{L}_{\text{abs+CO}} = 0.5\,\mathcal{L}_{\text{abs}} + 0.5\,(1 - \text{CO}) \tag{18}$$

I. PIB + Hard TP (quadratic penalty, $\lambda = 10$):

$$\mathcal{L}_{\text{hardTP}} = \mathcal{L}_{\text{PIB}} + 10 \cdot [\max(1 - \text{TP},\ 0)]^2 \tag{19}$$

Rationale: Quadratic penalty creates a steeper gradient near TP < 1, preventing gradual throughput erosion during training.
J. TP-Penalized PIB with weight sweep ($w \in \{0.5, 2.0\}$):

$$\mathcal{L}_w = \mathcal{L}_{\text{PIB}} + w \cdot \max(1 - \text{TP},\ 0) \tag{20}$$

K. Raw Received Power (no vacuum reference):

$$\mathcal{L}_{\text{rawRP}} = -\log\left(\frac{\sum_{\mathbf{r}\in\mathcal{B}} |U_{f,\text{pred}}(\mathbf{r})|^2}{\sum |U_{\text{in}}|^2} + \epsilon\right) \tag{21}$$

Physics rationale: This is the *most direct* formulation — it maximizes the focal-plane throughput into the fiber bucket without any reference to the vacuum field. The logarithm provides scale-invariant gradients.

5.2 Results: Convergence to a Physical Attractor

Table 5: 0403 results — 5 strategies on pitch_rescale data (100 test samples).

Strategy	PIB@10μm	TP	RP@10μm	ΔRP	TP σ
Vacuum	0.960	0.977	277.2	—	0.000
Turbulent	0.771	0.970	212.0	baseline	0.008
AbsBucket+CO	0.805	0.983	224.3	+5.8%	0.004
PIB+HardTP(λ=10)	0.804	**0.989**	225.0	+6.1%	**0.002**
TP-PIB w=0.5	**0.813**	0.969	223.8	+5.5%	0.008
TP-PIB w=2.0	0.804	**0.989**	225.0	+6.1%	**0.002**
Raw RP	0.808	0.986	**225.5**	+6.4%	0.003

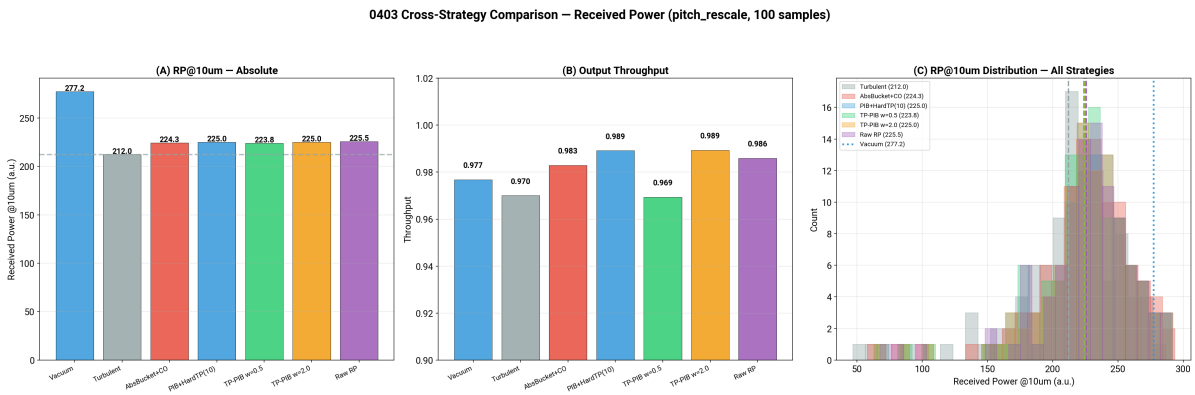


Figure 4: 0403 cross-strategy comparison. All TP-preserving losses exceed the turbulent baseline.

5.3 Per-Loss-Function Insights

5.3.1 Raw Received Power (Best: +6.4%)

The raw RP loss (Eq. 21) achieves the highest received power because it directly optimizes the quantity of interest without intermediate proxies. Key properties:

- No vacuum reference $\Rightarrow$ no ratio artifacts when vacuum is much higher than achievable
- Log-scale gradient: $\frac{\partial \mathcal{L}}{\partial \mathrm{RP}} \propto -1/\mathrm{RP}$, providing stronger gradients when RP is low
- Inherently throughput-preserving: scattering energy *reduces* the numerator

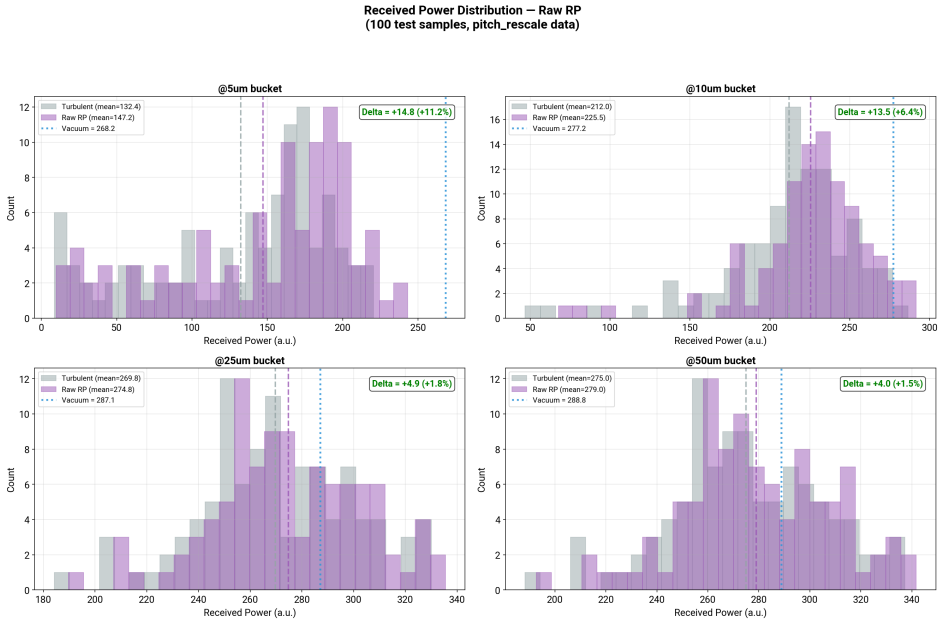


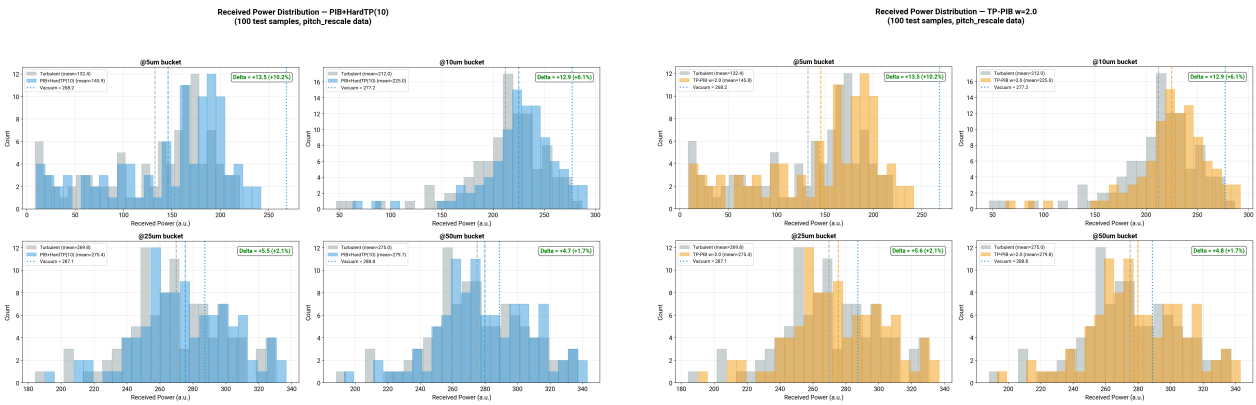
Figure 5: Raw RP: received power histogram showing consistent improvement across all bucket radii.

5.3.2 PIB + Hard TP and TP-PIB w=2.0 (Tied: +6.1%)

Both achieve identical performance, revealing that:

$$\mathcal{L}_{\text{PIB}} + 10\,[\max(1-\text{TP},0)]^2 \;\approx\; \mathcal{L}_{\text{PIB}} + 2.0\,\max(1-\text{TP},0) \tag{22}$$

at the operating point TP ≈ 0.99. The quadratic and strong linear penalties produce equivalent constraint forces near TP $= 1$. Both achieve the **lowest throughput variance** ($\sigma = 0.002$), indicating the most stable training dynamics.



(a) PIB + Hard TP ($\lambda = 10$) (b) TP-PIB $w = 2.0$

Figure 6: Nearly identical histograms confirm equivalent constraint behavior.

5.3.3 TP-PIB w=0.5 (Lowest: +5.5%)

The weakest TP constraint allows the D2NN to trade more throughput for PIB:

- Highest PIB@10μm (0.813) among all strategies
- But lowest TP (0.969) and highest TP variance ($\sigma = 0.008$)
- Net RP is lowest because the TP reduction outweighs the PIB gain

This confirms the relationship in Eq. 8: **TP preservation is more impactful than PIB maximization** for received power.

5.3.4 Absolute Bucket + CO (+5.8%)

Adding CO to the absolute bucket loss provides marginal wavefront correction:

- CO improves from 0.268 (baseline) to 0.270 (+0.7%)
- PIB@10μm: 0.805 (slightly lower than pure absolute bucket 0.814)
- The CO term competes with the bucket term, slightly reducing RP

Conclusion: At TP $\approx$ 98%, adding CO provides negligible benefit and slightly harms received power.

6 Wavefront Analysis: Why WF RMS Does Not Improve

A striking result across all TP-preserving strategies: **WF RMS remains essentially unchanged** ($\sim$442 nm, $<$ 0.3% change from baseline).

Table 6: Zernike decomposition summary (100 samples, 22 modes).

Strategy	Total RMS (turb)	Total RMS (D2NN)	Δ
AbsBucket+CO	362.8 nm	364.3 nm	+1.5 nm
PIB+HardTP	362.8 nm	363.3 nm	+0.6 nm
TP-PIB w=0.5	362.8 nm	364.8 nm	+2.1 nm
TP-PIB w=2.0	362.8 nm	363.3 nm	+0.5 nm
Raw RP	362.8 nm	363.9 nm	+1.1 nm

Physical interpretation: The D2NN is *not* performing wavefront correction. Instead, it redistributes the focal-plane energy pattern via deterministic diffraction, concentrating slightly more power into the central lobe without altering the wavefront. This is consistent with the theoretical limit: a static phase mask cannot conjugate a random, time-varying wavefront aberration.

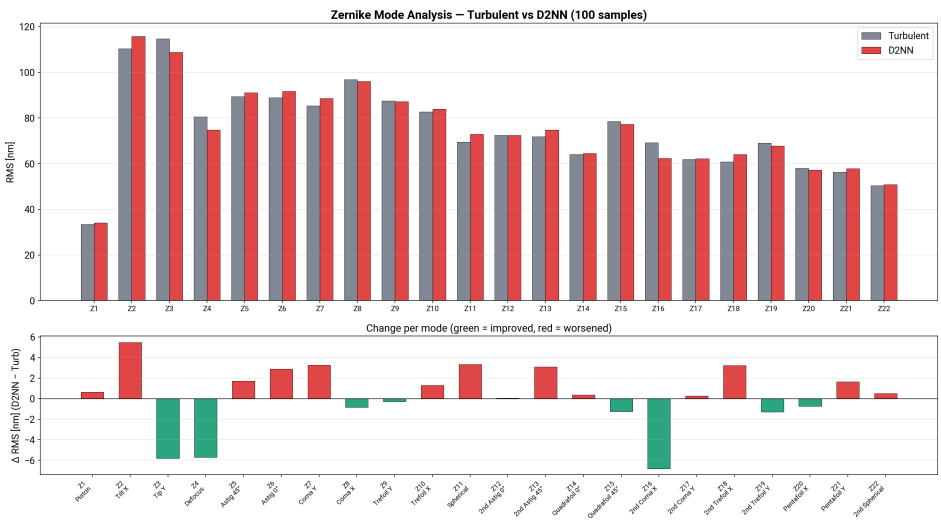


Figure 7: Zernike mode change for Raw RP: no significant correction at any mode.

7 Phase Mask Analysis

The learned phase masks for all TP-preserving strategies show similar structure:

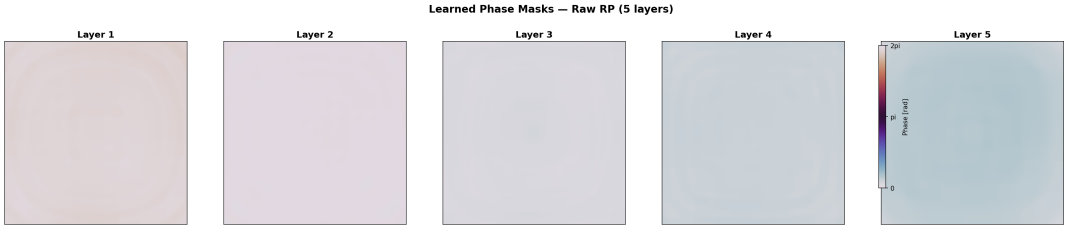


Figure 8: Phase masks for Raw RP loss. Predominantly low-spatial-frequency structure, no sharp gratings.

The absence of high-frequency grating structures confirms that the TP-preserving losses do not exploit the scattering mechanism observed in 0401. Instead, the gentle, low-frequency phase profiles suggest the D2NN acts as a weak Fresnel lens or defocus corrector that redistributes the focal intensity without significant energy loss.

8 Physical Mechanism: Focal Energy Redistribution

The +6% received power improvement arises from a subtle mechanism. Consider the turbulent focal field as a superposition of a coherent core and a speckle halo:

$$U_f = U_{\text{core}} + U_{\text{halo}} \tag{23}$$

The D2NN modifies the pupil-plane field such that after the focusing lens:

$$U'_f = U'_{\text{core}} + U'_{\text{halo}} \tag{24}$$

where $|U'_{\text{core}}|^2 > |U_{\text{core}}|^2$ within r_b, at the expense of $|U'_{\text{halo}}|^2$ being redistributed to larger radii (but still within the focal-plane window, not lost).

Since TP ≈ 0.99, the total energy is conserved. The D2NN effectively applies a weak spatial filter that tightens the PSF core by a few percent, which is sufficient to push more energy inside the $10\,\mu$m bucket.

This is not wavefront correction (which would require matching the turbulent phase), but rather a **statistical mode-filtering** operation: the static phase masks are optimized over the ensemble of turbulent realizations to find a configuration that, on average, redirects a small fraction of the halo energy back into the core.

9 Summary of All Loss Functions

Table 7: Complete loss function comparison across all sweeps.

Loss	Formula	PIB	TP	ΔRP	Verdict
PIB Only	$1 - \text{PIB}$	0.91	0.50	−41%	Fail
Strehl	$1 - S$	0.82	0.57	−40%	Fail
Intensity OL	$1 - \text{IO}$	0.26	0.08	−97%	Fail
CO+PIB	$(1-\text{CO}) + 0.5\mathcal{L}_{\text{PIB}}$	0.96	0.02	−98%	Fail
Multiplane	$0.5(1-\text{CO})+0.3\mathcal{L}_{\text{PIB}}+0.2\|A\|^2$	0.83	0.40	−55%	Fail
Abs. Bucket	$1 - E_b^{\text{pred}}/E_b^{\text{vac}}$	0.80	0.99	+5.9%	Pass
TP-PIB w=0.5	$\mathcal{L}_{\text{PIB}} + 0.5\,\text{TP}_\ell$	0.81	0.97	+5.5%	Pass
TP-PIB w=1.0	$\mathcal{L}_{\text{PIB}} + 1.0\,\text{TP}_\ell$	0.80	0.98	+5.8%	Pass
TP-PIB w=2.0	$\mathcal{L}_{\text{PIB}} + 2.0\,\text{TP}_\ell$	0.80	0.99	+6.1%	Pass
PIB+HardTP	$\mathcal{L}_{\text{PIB}} + 10(\text{TP}_\ell)^2$	0.80	0.99	+6.1%	Pass
AbsBkt+CO	$0.5\mathcal{L}_{\text{abs}} + 0.5(1-\text{CO})$	0.81	0.98	+5.8%	Pass
Raw RP	$-\log(E_b/E_{\text{in}})$	0.81	0.99	**+6.4%**	**Best**

10 Conclusions

1. **PIB is misleading.** Any loss function that only maximizes the power-in-bucket *ratio* will exploit the energy-scattering loophole, achieving high PIB at the cost of catastrophic throughput loss and net received power degradation.

2. **Throughput preservation is essential.** Only loss functions that explicitly constrain or inherently preserve throughput can achieve genuine received power improvement. The critical threshold is TP $> 95\%$.

3. **Static D2NN can improve received power by $\sim$6%.** For 1 km propagation with $C_n^2 = 5 \times 10^{-14}$, a 5-layer phase-only D2NN increases the absolute focal-plane received power within a 10 μm bucket by 5.5–6.4% over the uncorrected turbulent beam.

4. **The mechanism is energy redistribution, not wavefront correction.** WF RMS and Zernike decomposition are unchanged. The D2NN applies a weak, ensemble-optimized spatial filter that tightens the PSF core by a few percent.

5. **Raw received power loss is optimal.** The direct formulation $\mathcal{L} = -\log(E_b/E_{\text{in}})$ achieves the best performance (+6.4%) with high throughput (98.6%) and low variance, without requiring vacuum reference fields.

6. **TP penalty weight has diminishing returns.** Beyond $w \geq 2.0$ (or $\lambda \geq 10$ quadratic), all strategies converge to the same $\sim$6% attractor, suggesting a fundamental physical limit of static phase-only D2NN for this turbulence regime.